Assumptions and Limitations

Brent Oil Change Point Analysis | Birhan Energies

Assumptions

- **Event dates are approximate "start dates"** for each entry in data/events.csv. Real-world events (wars, sanctions regimes, OPEC policy shifts) unfold over days to months; a single date is a simplification used to align events with the daily price series.

- **Brent price is treated as a single global benchmark.** We do not adjust for inflation, currency effects (all prices are nominal USD/barrel), or the fact that Brent, WTI, and OPEC basket prices can diverge during regional supply shocks.

- **A mean-shift change point model is an adequate first approximation.** We assume that, locally around a structural break, the price process can be summarised by a constant mean plus noise. This is a simplification - markets also exhibit trends, mean-reversion, and volatility regimes that a single-mean-shift model does not capture.

- **One (or a small, fixed number of) change point(s) are estimated at a time** for interpretability. In reality, the 35-year series plausibly contains dozens of regime shifts; the model is extended incrementally rather than fit with an unbounded number of change points in the first pass.

- **The trading calendar** (holidays, weekends, exchange closures) is not explicitly modeled. Gaps between consecutive trading days are treated as if they were of equal length.

Limitations

- **Correlation in time is not proof of causation.** This is the central methodological caveat of the whole analysis and is expanded on below.

- **Data does not capture magnitude of an event.** data/events.csv records that an event occurred and a qualitative expected direction, not a quantitative measure of its severity, so the strength of any detected shift cannot be directly attributed to "how big" the event was.

- **Multiple plausible causes can coincide.** Oil markets are influenced by simultaneous macro factors (currency moves, demand shocks, OPEC policy, and geopolitical events can overlap within days of each other), which makes it difficult to isolate a single "cause" for an inferred change point.

- **Change point models detect breaks in the modeled parameter only** (e.g. the mean). They do not, by themselves, explain why the break occurred - that step always requires the analyst to bring in outside context (the event list) and reason qualitatively about plausibility and timing.

- **Look-ahead / hindsight bias risk.** Because the event list is compiled with the benefit of hindsight, there is a risk of confirmation bias when matching change points to events. We mitigate this by requiring temporal proximity (the event should precede or coincide with, not follow, the detected shift) and by being explicit that these are hypotheses, not proven causal claims.

Correlation vs. Causal Impact

A Bayesian change point model identifies a date (or a posterior distribution over dates) at which the statistical properties of the price series most likely shifted, and it quantifies that shift (e.g., the change in mean price). This is fundamentally a **descriptive, correlational** result: it tells us when the data-generating process changed and by how much.

It does **not**, on its own, establish that a specific real-world event caused that shift. To move from correlation to a causal claim one would need, at minimum:

- **Temporal precedence** - the event must precede the change point, not follow it (we check this, but proximity alone is weak evidence).

- **A plausible mechanism** - an economic/geopolitical channel by which the event could move the balance of oil supply, demand, or risk premium.

- **Ruling out confounders** - other events, seasonal effects, or macroeconomic shifts occurring in the same window that could equally explain the change.

- **Counterfactual reasoning or a natural-experiment design** (e.g. difference-in-differences, synthetic control, or event-study methods with a control series) to estimate what prices would have done absent the event - something a single-series change point model cannot provide.

Throughout this project we therefore frame event associations as **hypotheses consistent with the evidence** ("the detected change point around [date] is temporally consistent with [event], and a price shift in the [direction] implied by the event's likely supply/demand effect"), not as proven causal statements.